

Inventory Management

Inventory Management

Dr Vinod Yadav,
Assistant Professor,
Manipal University Jaipur.

What Is Inventory?

- Stock of items kept to meet future demand
- Purpose of inventory management
 - how many units to order
 - when to order

Supply Chain Management

- Bullwhip effect
 - demand information is distorted as it moves away from the end-use customer
 - higher safety stock inventories are stored to compensate
- Seasonal or cyclical demand
- Inventory provides independence from vendors
- Take advantage of price discounts
- Inventory provides independence between stages and avoids work stoppages

Quality Management in the Supply Chain

- Customers usually perceive quality service as availability of goods they want when they want them
- Inventory must be sufficient to provide high-quality customer service in QM

Forms of Inventory

- Raw materials
- Purchased parts and supplies
- Work-in-process (partially completed) products (WIP)
- Items being transported
- Tools and equipment

Two Forms of Demand

- Dependent
 - Demand for items used to produce final products
 - Tires for autos are a dependent demand item
- Independent
 - Demand for items used by external customers
 - Cars, appliances, computers, and houses are examples of independent demand inventory

Inventory Costs

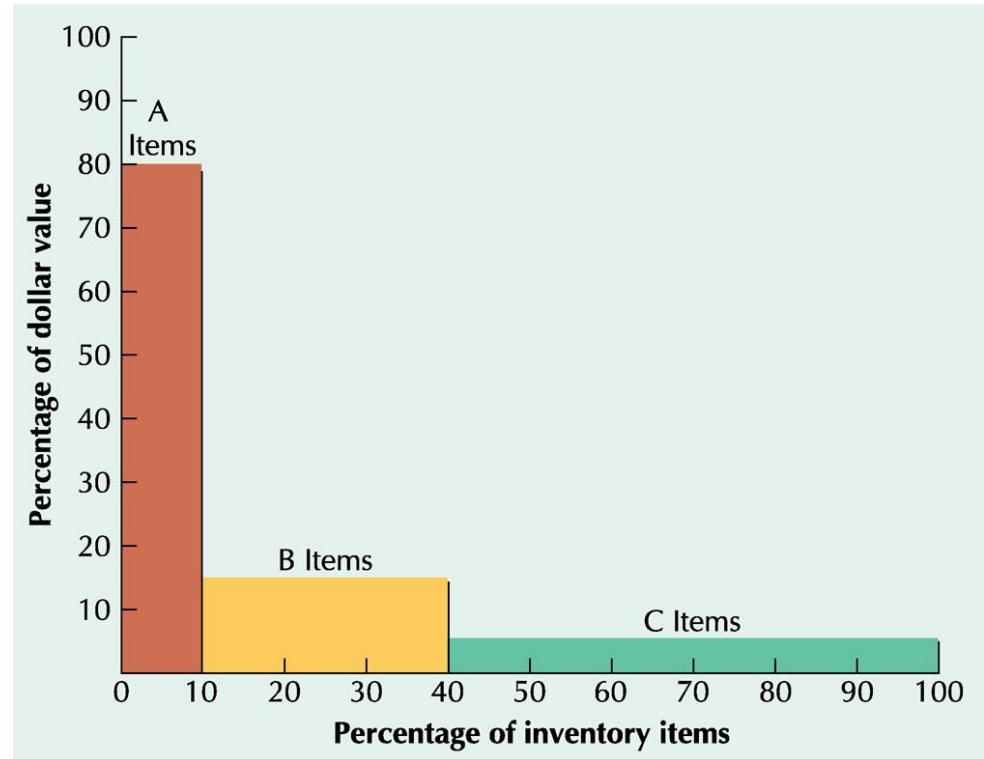
- Carrying cost
 - cost of holding an item in inventory
- Ordering cost
 - cost of replenishing inventory
- Shortage cost
 - temporary or permanent loss of sales when demand cannot be met

Inventory Control Systems

- Continuous system (fixed-order-quantity)
 - constant amount ordered when inventory declines to predetermined level
- Periodic system (fixed-time-period)
 - order placed for variable amount after fixed passage of time

The ABC Classification System

- Class A
 - 5 – 15 % of units
 - 70 – 80 % of value
- Class B
 - 30 % of units
 - 15 % of value
- Class C
 - 50 – 60 % of units
 - 5 – 10 % of value



ABC System Classification

PART	UNIT COST	ANNUAL USAGE
------	-----------	--------------

1	\$ 60	90
---	-------	----

2	350	40
---	-----	----

3	30	130
---	----	-----

4	80	60
---	----	----

5	30	100
---	----	-----

6	20	180
---	----	-----

7	10	170
---	----	-----

8	320	50
---	-----	----

9	510	60
---	-----	----

10	20	120
----	----	-----

ABC System Classification

PART	TOTAL VALUE	% OF TOTAL VALUE	% OF TOTAL QUANTITY	% CUMMULATIVE	
9	\$30,600	35.9	6.0	6.0	A
8	16,000	18.7	5.0	11.0	
2	14,000	16.4	4.0	15.0	
1	5,400	6.3	9.0	24.0	B
4	4,800	5.6	6.0	30.0	
3	3,900	4.6	13.0	43.0	
6	3,600	4.2	18.0	61.0	C
5	3,000	3.5	13.0	71.0	
10	2,400	2.8	12.0	83.0	
7	1,700	2.0	17.0	100.0	
\$85,400					

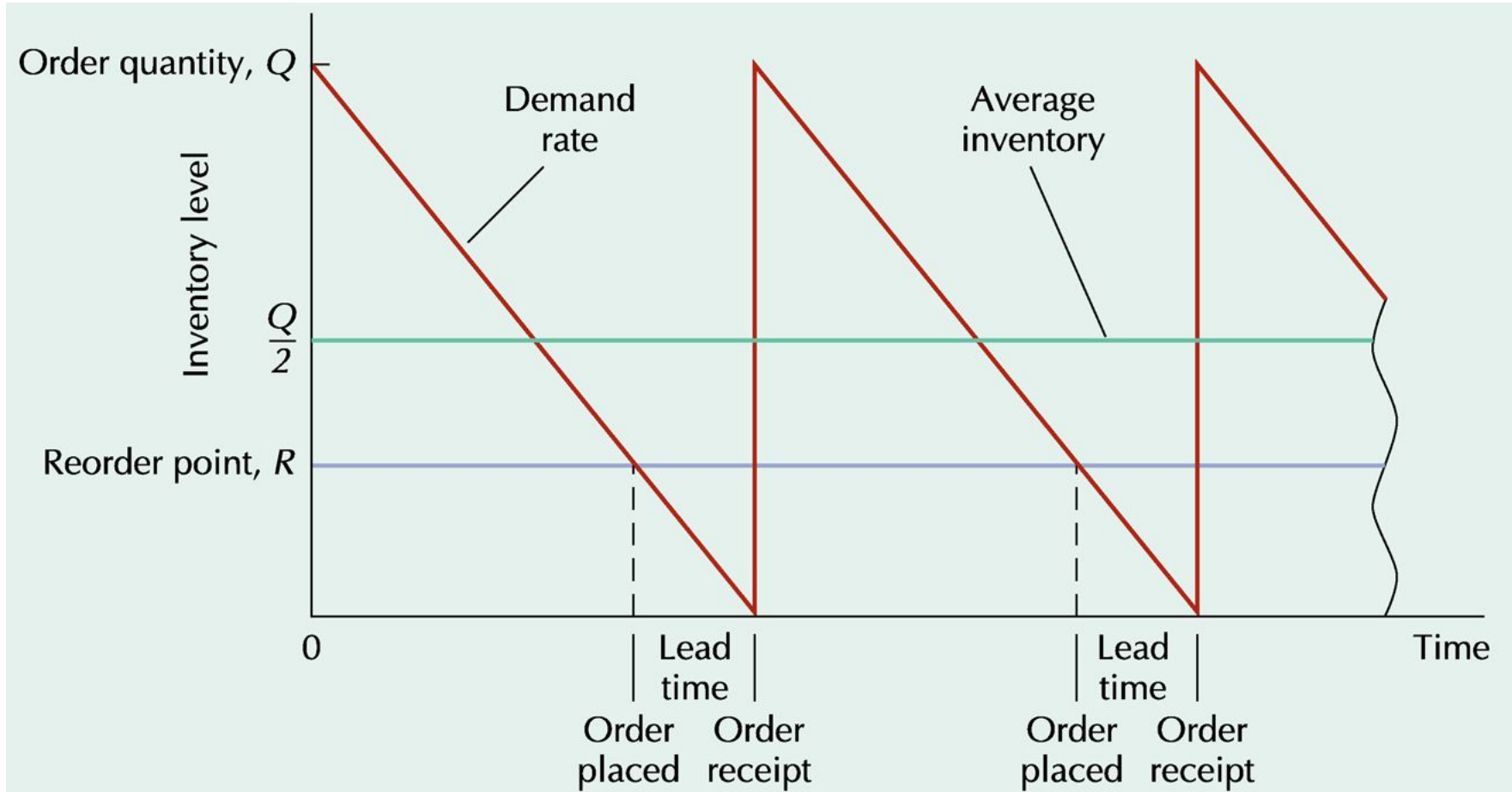
ABC System Classification

		% OF TOTAL CLASS ITEMS	VALUE	% OF TOTAL QUANTITY
A	9, 8, 2	71.0	15.0	
B	1, 4, 3	16.5	28.0	
C	6, 5, 10, 7	12.5	60.0	

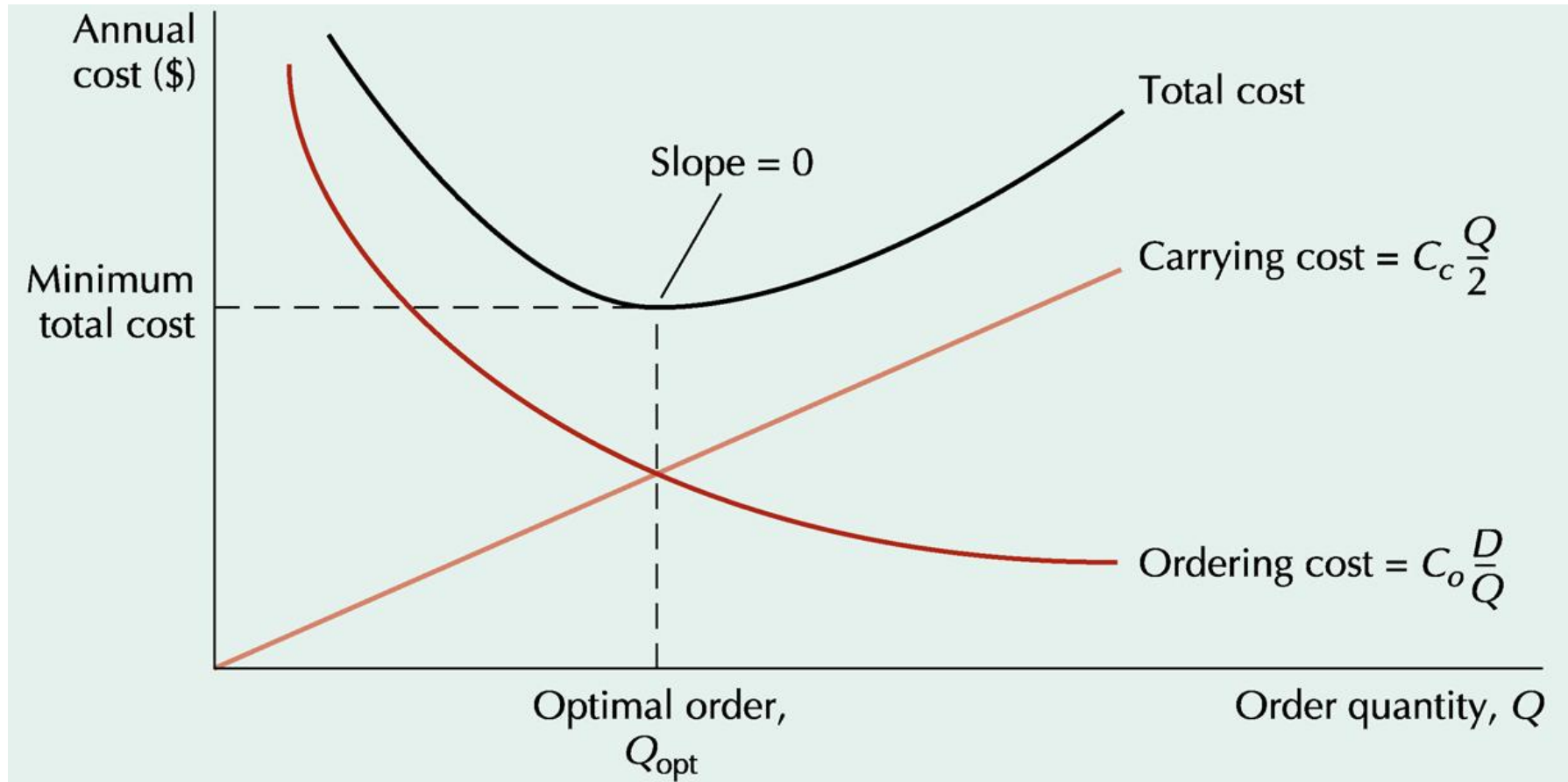
Economic Order Quantity (EOQ) Models

- EOQ
 - continuous inventory system
 - optimal order quantity that will minimize total inventory costs
- Basic EOQ model
 - Demand is known with certainty and is constant over time
 - No shortages are allowed
 - Lead time for the receipt of orders is constant
 - Order quantity is received all at once
- Production quantity model
 - Order is received gradually, as inventory is simultaneously being depleted
 - non-instantaneous receipt model
 - assumption that Q is received all at once is relaxed
 - p - daily rate at which an order is received over time, the *production rate*
 - d - daily rate at which inventory is demanded
- Order cycle
 - the time between receipt of orders in an inventory system

Inventory Order Cycle



EOQ Cost Model



EOQ Cost Model

C_o - cost of placing order D - annual demand
 C_c - annual per-unit carrying cost Q - order quantity

$$\text{Annual ordering cost} = \frac{C_o D}{Q}$$

$$\text{Annual carrying cost} = \frac{C_c Q}{2}$$

$$\text{Total cost} = \frac{C_o D}{Q} + \frac{C_c Q}{2}$$

EOQ Cost Model

Deriving Q_{opt}

$$TC = \frac{C_o D}{Q} + \frac{C_c Q}{2}$$

$$\frac{\partial TC}{\partial Q} = -\frac{C_o D}{Q^2} + \frac{C_c}{2}$$

$$0 = -\frac{C_o D}{Q^2} + \frac{C_c}{2}$$

$$Q_{\text{opt}} = \sqrt{\frac{2C_o D}{C_c}}$$

Proving equality of costs at optimal point

$$\frac{C_o D}{Q} = \frac{C_c Q}{2}$$

$$Q^2 = \frac{2C_o D}{C_c}$$

$$Q_{\text{opt}} = \sqrt{\frac{2C_o D}{C_c}}$$

EOQ Example

$$C_c = \$0.75 \text{ per gallon} \quad C_o = \$150 \quad D = 10,000 \text{ gallons}$$

$$Q_{\text{opt}} = \sqrt{\frac{2C_o D}{C_c}}$$

$$TC_{\text{min}} = \frac{C_o D}{Q} + \frac{C_c Q}{2}$$

$$Q_{\text{opt}} =$$

$$TC_{\text{min}} =$$

$$Q_{\text{opt}} =$$

$$TC_{\text{min}} =$$

Orders per year =

Order cycle time

$$D/Q_{\text{opt}}$$

=

EOQ Example

$$C_c = \$0.75 \text{ per gallon} \quad C_o = \$150 \quad D = 10,000 \text{ gallons}$$

$$Q_{\text{opt}} = \sqrt{\frac{2C_o D}{C_c}}$$

$$TC_{\text{min}} = \frac{C_o D}{Q} + \frac{C_c Q}{2}$$

$$Q_{\text{opt}} = \sqrt{\frac{2(150)(10,000)}{(0.75)}}$$

$$TC_{\text{min}} = \frac{(150)(10,000)}{2,000} + \frac{(0.75)(2,000)}{2}$$

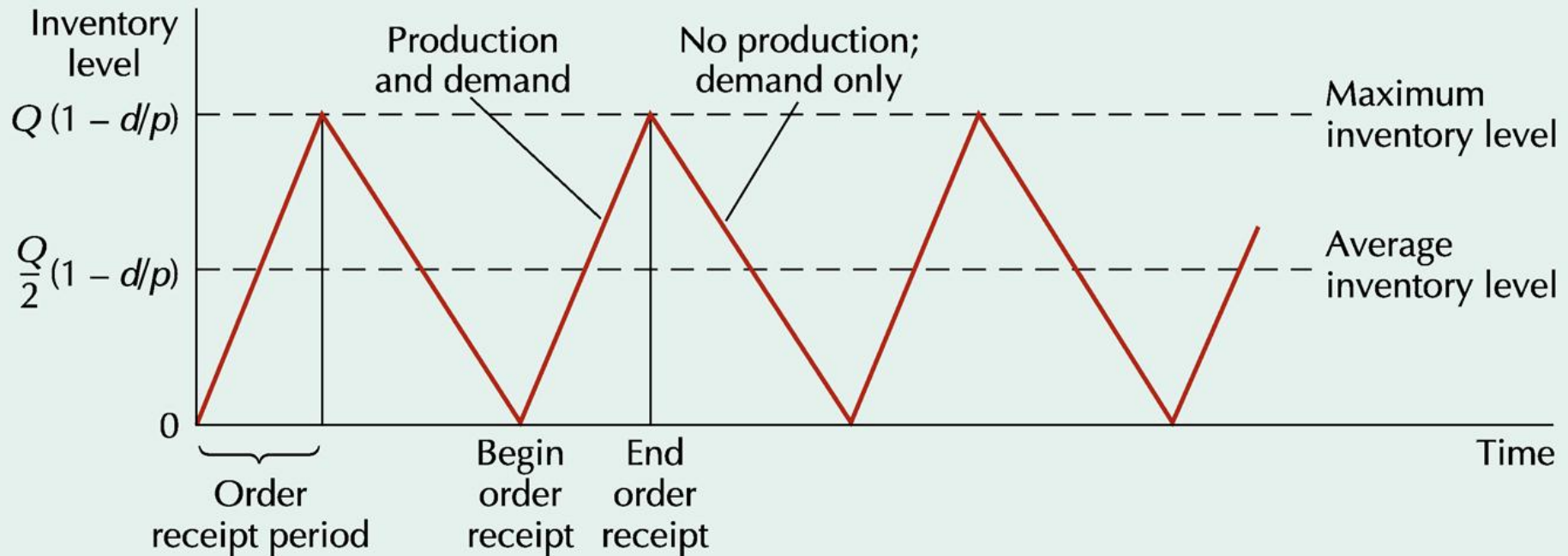
$$Q_{\text{opt}} = 2,000 \text{ gallons}$$

$$TC_{\text{min}} = \$750 + \$750 = \$1,500$$

$$\begin{aligned} \text{Orders per year} &= D/Q_{\text{opt}} \\ &= 10,000/2,000 \\ &= 5 \text{ orders/year} \end{aligned}$$

$$\begin{aligned} \text{Order cycle time} &= 311 \\ &\text{days}/(D/Q_{\text{opt}}) \\ &= 311/5 \\ &= 62.2 \text{ store days} \end{aligned}$$

Production Quantity Model



Production Quantity Model

p = production rate d = demand rate

$$\begin{aligned} \text{Maximum inventory level} &= \frac{Q}{p} \\ d & \\ &= Q \left(1 - \frac{d}{p} \right) \end{aligned}$$

$$\text{Average inventory level} = \frac{Q}{2} \left(1 - \frac{d}{p} \right)$$

$$Q_{\text{opt}} = \sqrt{\frac{2C_o D}{C_c \left(1 - \frac{d}{p} \right)}}$$

$$TC = \frac{C_o D}{Q} + \frac{C_c Q}{2} \left(1 - \frac{d}{p} \right)$$

Production Quantity Model

$C_c = \$0.75$ per gallon $C_o = \$150$ $D = 10,000$ gallons
 $d = 10,000/311 = 32.2$ gallons per day $p = 150$ gallons per day

$$Q_{\text{opt}} = \sqrt{\frac{2C_o D}{C_c \left(1 - \frac{d}{p}\right)}} =$$

$$TC = \frac{C_o D}{Q} + \frac{C_c Q}{2} \left(1 - \frac{d}{p}\right) =$$

$$\text{Production run} = \frac{Q}{p} =$$

Production Quantity Model

$$\text{Number of production runs} = \frac{D}{Q} =$$

$$= \text{Maximum inventory level} = Q \left(1 - \frac{d}{p} \right)$$

Production Quantity Model

$C_c = \$0.75$ per gallon $C_o = \$150$ $D = 10,000$ gallons
 $d = 10,000/311 = 32.2$ gallons per day $p = 150$ gallons per day

$$Q_{\text{opt}} = \sqrt{\frac{2C_o D}{C_c \left(1 - \frac{d}{p}\right)}} = \sqrt{\frac{2(150)(10,000)}{0.75 \left(1 - \frac{32.2}{150}\right)}} = 2,256.8 \text{ gallons}$$

$$TC = \frac{C_o D}{Q} + \frac{C_c Q}{2} \left(1 - \frac{d}{p}\right) =$$

\$1,329

$$\text{Production run} = \frac{Q}{p} = \frac{2,256.8}{150} = 15.05 \text{ days per order}$$

Production Quantity Model

$$\text{Number of production runs} = \frac{D}{Q} = \frac{10,000}{2,256.8} = 4.43 \text{ runs/year}$$

$$\begin{aligned} 1 - \text{Maximum inventory level} &= Q \left(1 - \frac{d}{p} \right) = 2,256.8 \left(1 - \frac{32.2}{150} \right) \\ &= 1,772 \text{ gallons} \end{aligned}$$

Solution of EOQ Models With Excel

D8		fx		=SQRT((2*D5*D6)/D4)	
	A	B	C	D	E
1	Example 13.2: The Economic Order Quantity				
2					
3					
4		Carrying cost =	\$	0.75	
5		Ordering cost =	\$	150	
6		Demand =		10,000	
7					
8		Q =		2,000	gallons
9		TC =	\$	1,500	
10		Order per year =		5	orders
11		Order cycle time =		62.20	days
12					

The optimal order size, Q, in cell D8

Solution of EOQ Models With Excel

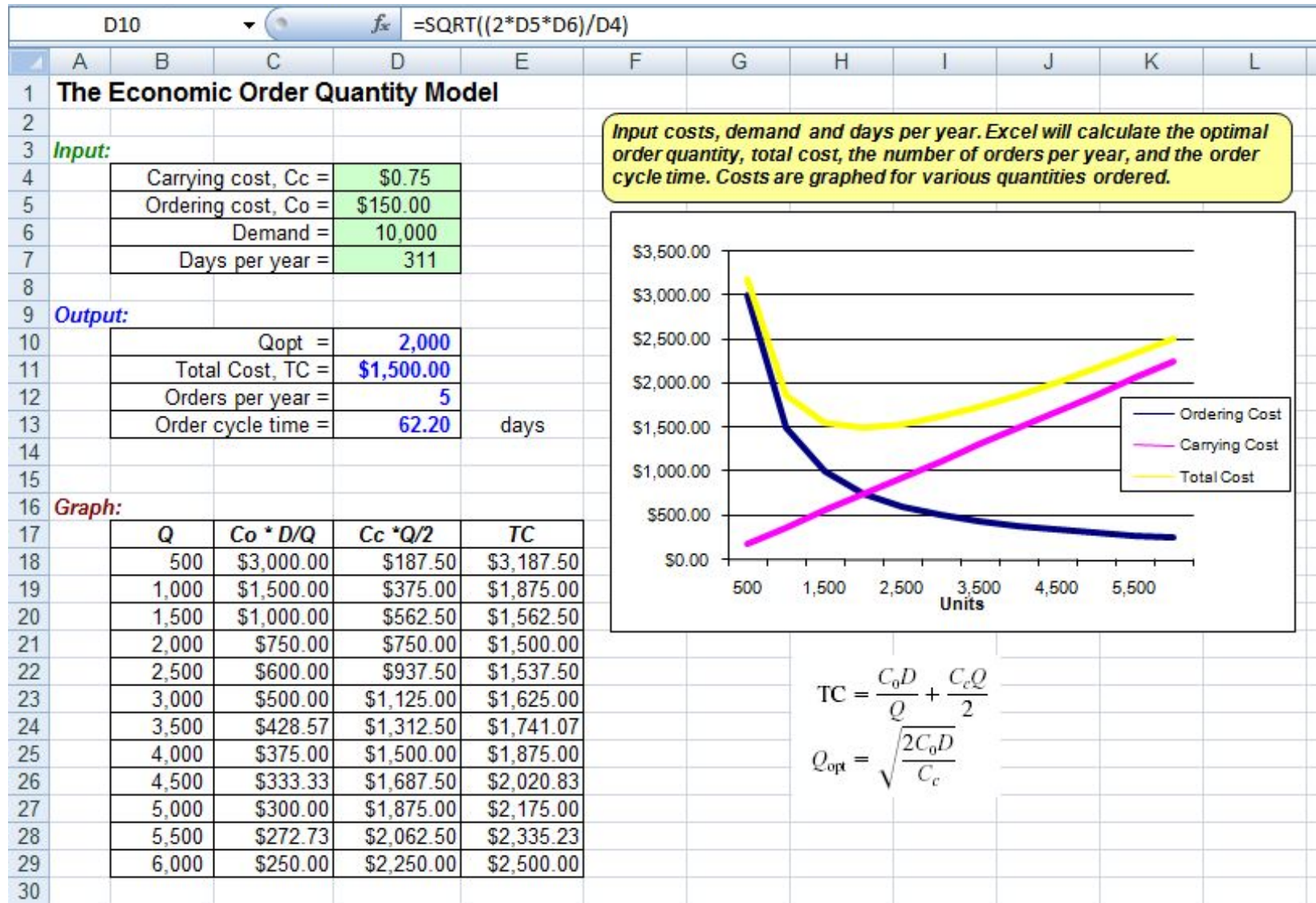
D10		fx		=SQRT(2*D4*D5/(D3*(1-(D7/D8))))		
	A	B	C	D	E	F
1	Example 13.3: The Production Quantity Model					
2						
3			Carrying cost = \$	0.75		
4			Ordering cost = \$	150		
5			Demand =	10,000		
6			Annual days =	311		
7			Daily demand rate =	32.15		
8			Daily production rate =	150		
9						
10			Q =	2256.41		
11			TC = \$	1329.54		
12			Production run length =	15.04		
13			Number of runs =	4.43		
14			Maximum inventory =	1772.72		

The formula for Q in cell D10

$=(D4*D5/D10)+(D3*D10/2)*(1-(D7/D8))$

$=D10*(1-(D7/D8))$

Solution of EOQ Models With OM Tools



Quantity Discounts

Price per unit decreases as order quantity increases

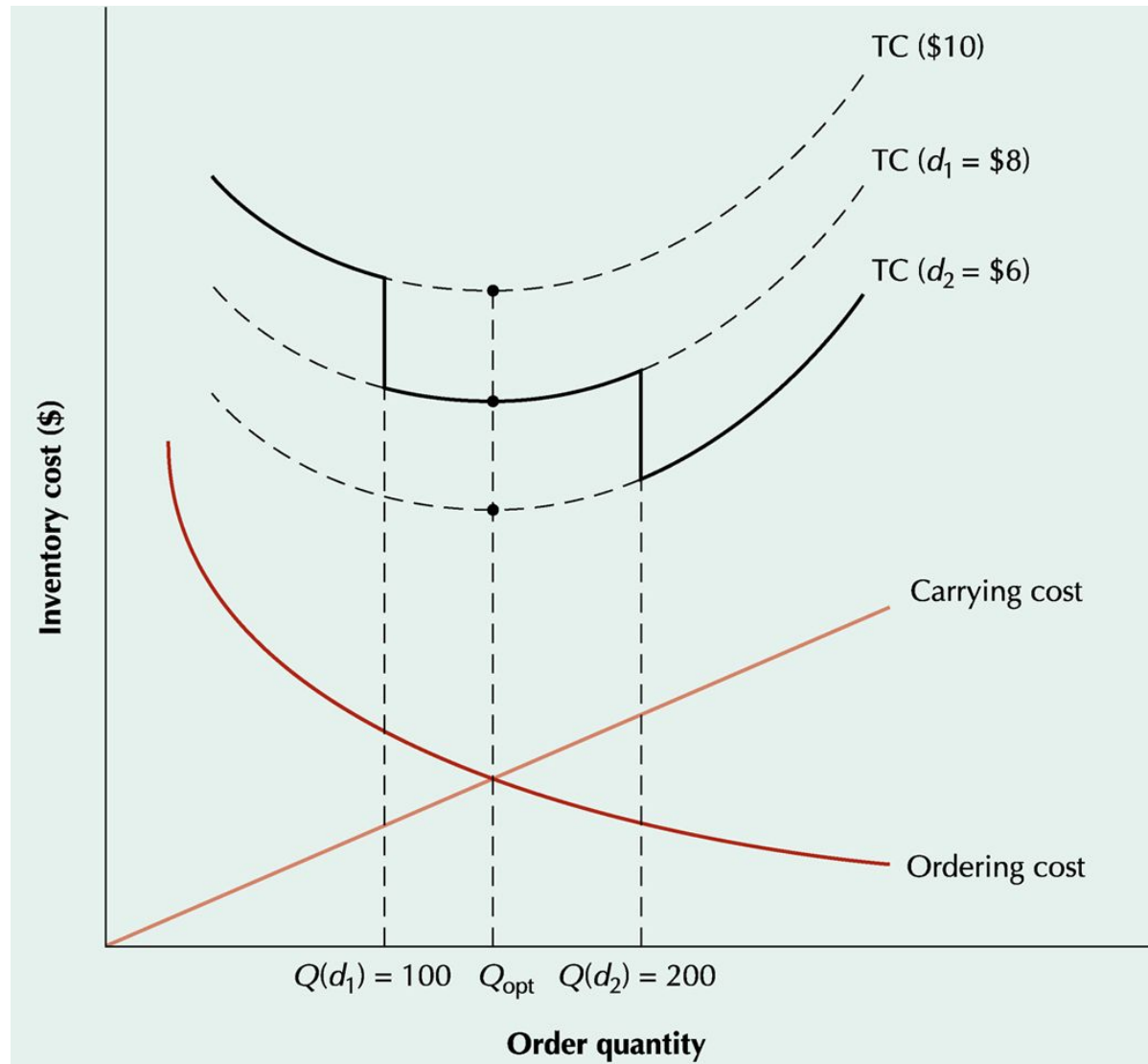
$$TC = \frac{C_o D}{Q} + \frac{C_c Q}{2} + PD$$

where

P = per unit price of the item

D = annual demand

Quantity Discounts with Constant Carrying Cost



Quantity Discount

QUANTITY PRICE

1 - 49 \$1,400

50 - 89 1,100

90+ 900

$C_o = \$2,500$

$C_c = \$190$ per

TV

$D = 200$ TVs per

year

$$Q_{\text{opt}} = \sqrt{\frac{2C_o D}{C_c}} =$$

For $Q =$

$$TC = \frac{C_o D}{Q_{\text{opt}}} + \frac{C_c Q_{\text{opt}}}{2} + PD =$$

For $Q = 90$

$$TC = \frac{C_o D}{Q} + \frac{C_c Q}{2} + PD =$$

Quantity Discount

QUANTITY PRICE

1 - 49 \$1,400

50 - 89 1,100

90+ 900

$C_o = \$2,500$

$C_c = \$190$ per

TV

$D = 200$ TVs per

$$Q_{\text{opt}} = \sqrt{\frac{2C_o D}{C_c}} = \sqrt{\frac{2(2500)(200)}{190}} = 72.5 \text{ TVs}$$

For $Q = 72.5$

$$TC = \frac{C_o D}{Q_{\text{opt}}} + \frac{C_c Q_{\text{opt}}}{2} + PD = \$233,784$$

For $Q = 90$

$$TC = \frac{C_o D}{Q} + \frac{C_c Q}{2} + PD =$$

\$194,105

Quantity Discount Model With Excel

E8		fx		=IF(D8>=B8,D8,B8)			
	A	B	C	D	E	F	G
1	Example 13.4: A Quantity Discount Model with Constant Carrying Cost						
2							
3			Carrying cost =	\$ 190			
4			Ordering cost =	\$ 2,500			
5			Demand =	200			
6							
7		Quantity	Price	Q	Discount Q	Total Cost	
8		1	1,400	72.55	72.55	\$ 293,784.05	
9		50	1,100	72.55	72.55	\$ 233,784.05	
10		90	900	72.55	90.00	\$ 194,105.56	<i>Optimal</i>
11							

=IF(D10>B10,D10,B10)

=(D4*D5/E10)+(D3*E10/2)+C10*D5

Reorder Point

- Inventory level at which a new order is placed

$$R = dL$$

where

d = demand rate per period

L = lead time

Reorder Point

Demand = 10,000 gallons/year

Store open 311 days/year

Daily demand =

Lead time = L = 10 days

$R = dL =$

Reorder Point

Demand = 10,000 gallons/year

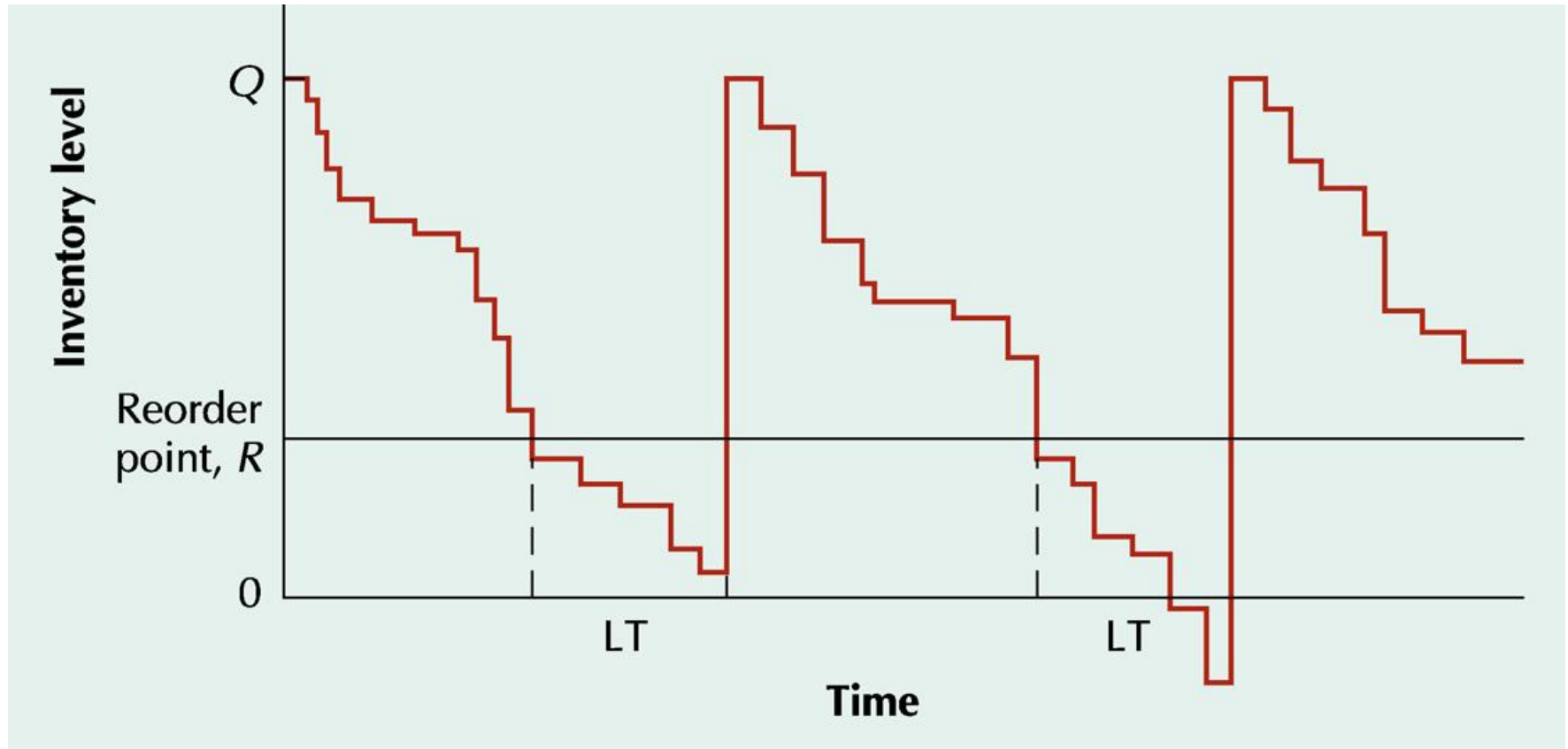
Store open 311 days/year

Daily demand = $10,000 / 311 = 32.154$
gallons/day

Lead time = $L = 10$ days

$R = dL = (32.154)(10) = 321.54$ gallons

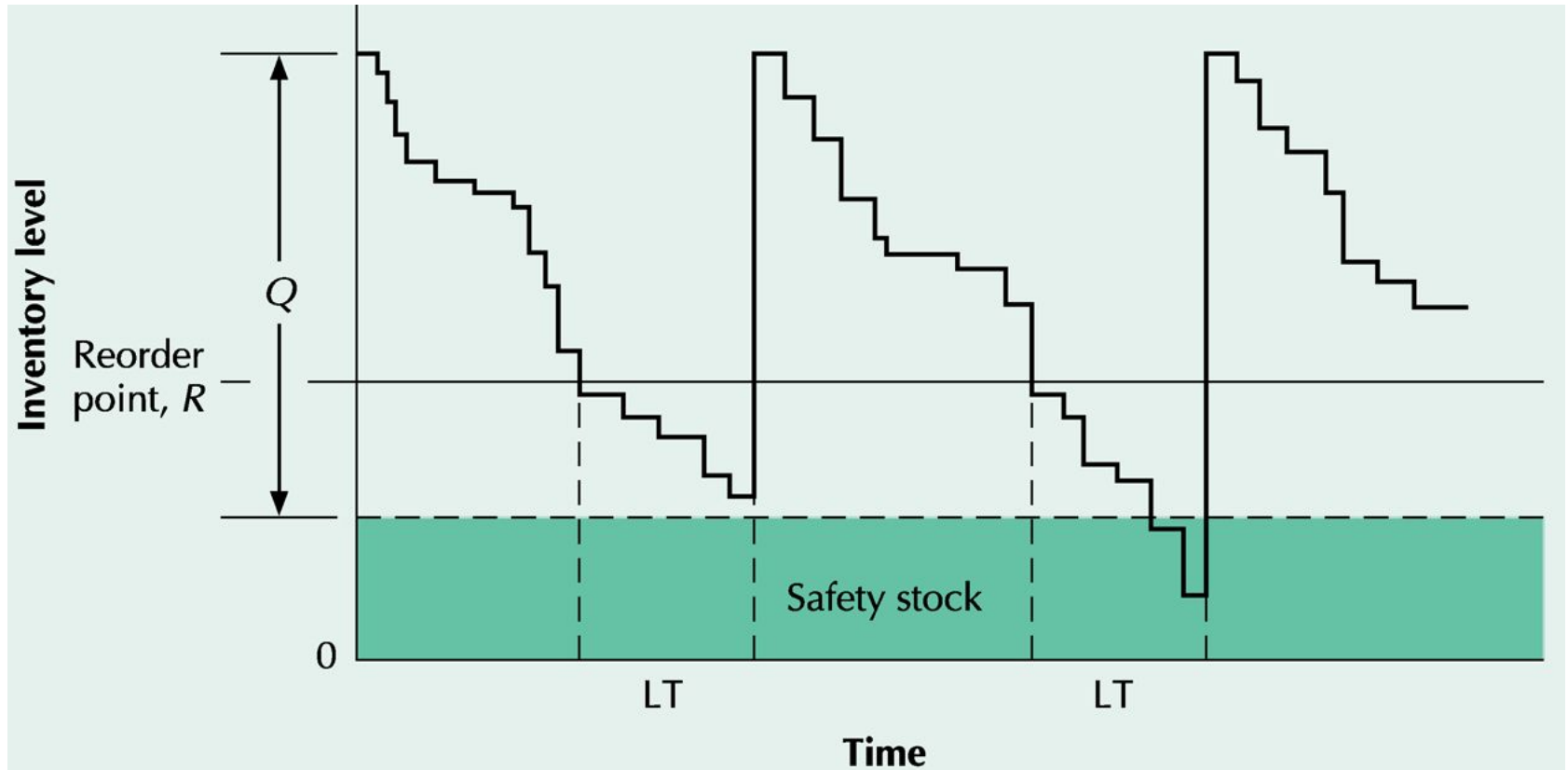
Variable Demand with a Reorder Point



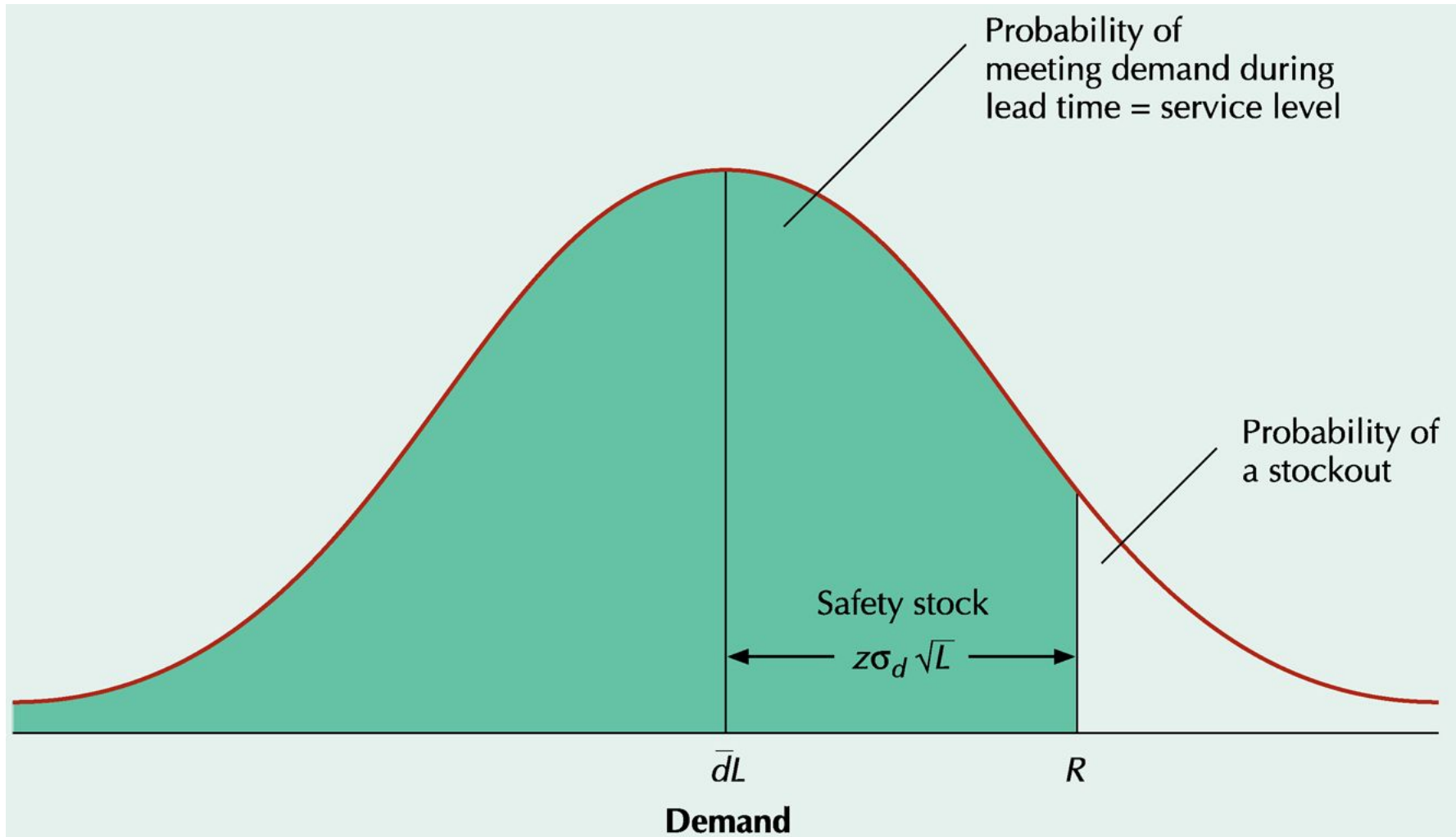
Safety Stock

- Safety stock
 - buffer added to on hand inventory during lead time
- Stockout
 - an inventory shortage
- Service level
 - probability that the inventory available during lead time will meet demand
 - $P(\text{Demand during lead time} \leq \text{Reorder Point})$

Reorder Point with a Safety Stock



Reorder Point For a Service Level



Reorder Point With Variable Demand

$$R = \bar{d}L + z\sigma_d \sqrt{L}$$

where

$\bar{d}$ = average daily demand

L = lead time

σ_d = the standard deviation of daily demand

z = number of standard deviations
corresponding to the service level
probability

$z\sigma_d \sqrt{L}$ = safety stock

Reorder Point For Variable Demand

The paint store wants a reorder point with a 95% service level and a 5% stockout probability

$$\bar{d} = 30 \text{ gallons per day}$$

$$L = 10 \text{ days}$$

$$\sigma_d = 5 \text{ gallons per day}$$

For a 95% service level, $z = 1.65$

$$R = \bar{d}L + z \sigma_d \sqrt{L}$$

$$\text{Safety stock} = z \sigma_d \sqrt{L}$$

Reorder Point For Variable Demand

The paint store wants a reorder point with a 95% service level and a 5% stockout probability

$$\bar{d} = 30 \text{ gallons per day}$$

$$L = 10 \text{ days}$$

$$\sigma = 5 \text{ gallons per day}$$

For a 95% service level, $z = 1.65$

$$R = \bar{d}L + z\sigma\sqrt{L}$$

$$= 30(10) + (1.65)(5)(\sqrt{10})$$

10)

$$= 326.1 \text{ gallons}$$

$$\text{Safety stock} = z\sigma\sqrt{L}$$

$$= (1.65)(5)(\sqrt{10})$$

$$= 26.1 \text{ gallons}$$

Determining Reorder Point with Excel

E7 fx =E3*E5+1.65*E4*SQRT(E5)						
	A	B	C	D	E	F
1	Example 13.6: Reorder Point with Variable Demand					
2						
3			Average daily demand =		30	
4			Standard deviation =		5	
5			Lead time =		10	
6						
7				$R =$	326.09	
8						

The reorder point
formula in cell E7

Order Quantity for a Periodic Inventory System

$$Q = \bar{d}(t_b + L) + z\sigma_d \sqrt{t_b + L} - I$$

where

$\bar{d}$ = average demand rate

t_b = the fixed time between orders

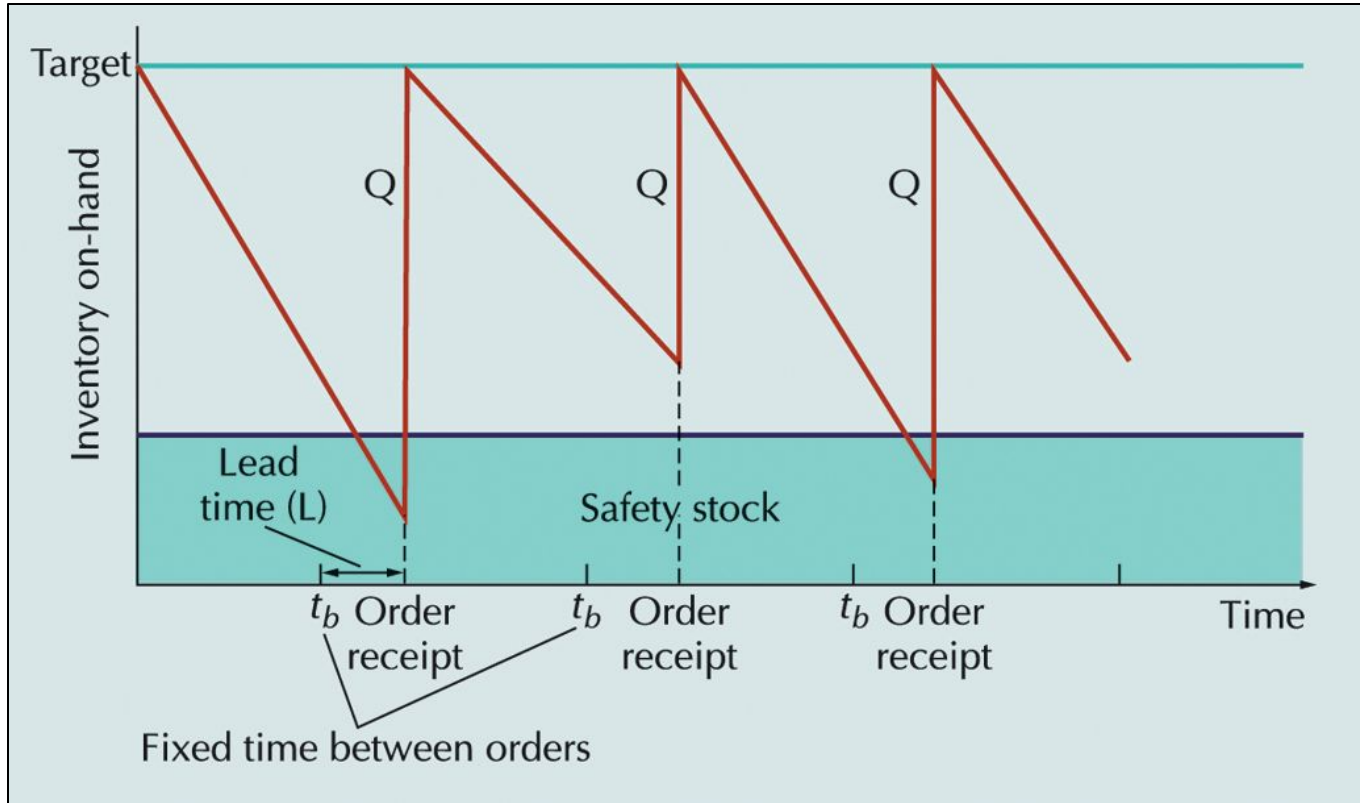
L = lead time

σ_d = standard deviation of demand

$z\sigma_d \sqrt{t_b + L}$ = safety stock

I = inventory level

Periodic Inventory System



Fixed-Period Model With Variable Demand

d = 6 packages per day

σ_d = 1.2 packages

t_b = 60 days

L = 5 days

I = 8 packages

z = 1.65 (for a 95% service level)

$$Q = d(t_b + L) + z\sigma_d \sqrt{t_b + L} - I$$

Fixed-Period Model With Variable Demand

d = 6 packages per day

σ_d = 1.2 packages

t_b = 60 days

L = 5 days

I = 8 packages

z = 1.65 (for a 95% service level)

$$Q = d(t_b + L) + z\sigma_d\sqrt{t_b + L} - I$$

$$= (6)(60 + 5) + (1.65)(1.2)\sqrt{60 + 5}$$

- 8

$$= 397.96 \text{ packages}$$

Fixed-Period Model with Excel

D10		fx		=D3*(D4+5)+D7-D8		
	A	B	C	D	E	F
1	Example 13.7: Fixed Period Model with Variable Demand					
2						
3			Average demand rate =	6	packages per day	
4			Time between orders =	60	days	
5			Lead time =	5	days	
6			Standard deviation of demand =	1.2	packages	
7			Safety stock =	15.96	packages	
8			Inventory in stock =	8	packages	
9						
10			Q =	397.96	packages	
11						

Formula for order size, Q, in cell D10